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Prebiotic magnetite enables chirality-magnetic surface feedback

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The emergence of biomolecular homochirality requires both an initial symmetry-breaking event and a mechanism to amplify and preserve a chiral imbalance. Magnetic minerals have been shown to function as chiral agents through the chiral-induced spin selectivity (CISS) effect and may have enabled homochirality on early Earth, yet the magnetic properties of magnetite formed under realistic prebiotic conditions remain unexplored. Here, we show that magnetite synthesized through two geochemically plausible pathways—UV-driven photo-oxidation and nitrite-mediated oxidation of Fe(II)—produces particles dominated by single-vortex and multivortex magnetic domain states. Magnetic measurements and electron microscopy confirm that these populations differ markedly from the nano-fabricated thin-film substrates conventionally used in previous CISS experiments. Using three-dimensional micromagnetic simulations, we demonstrate that single-domain and vortex-state grains undergo irreversible, exchange-driven remagnetization when interacting with spin-polarized homochiral compounds. This magnetic irreversibility provides a robust mechanism for storing and reinforcing weak chiral bias, suggesting that prebiotic magnetite could have contributed to the emergence and stabilization of persistent chiral bias on early Earth.

INTRODUCTION

The origin of life's homochirality remains one of the central challenges in understanding life's emergence. The uniform handedness of biomolecules underlies the structure and function of biopolymers such as RNA, proteins, and phospholipids, and is therefore considered an essential signature of life (*1*–*3*). To enable the high-yielding synthesis of functional polymers, homochirality must have been established before the onset of biological replication, through a sequence involving an initial symmetry-breaking mechanism, followed by the amplification and propagation of an induced enantiomeric bias, ultimately yielding a self-sustaining homochiral network (*4*–*6*). Understanding how life's homochirality emerged therefore requires identifying the natural planetary environments on early Earth that could have provided the conditions necessary to achieve such an outcome (*7*).

Various mechanisms have been proposed as potential pathways for the origin of homochirality, including parity-violating energy differences, circularly polarized light, asymmetric autocatalysis, and polarized particles produced by cosmic-ray showers, to name a few (*2*, *8*–*15*). Yet, these processes are generally considered insufficient to account for life's homochirality, as they generate only small enantiomeric excesses and often operate under conditions unlikely to have been present on early Earth. Moreover, they lack both an effective mechanism to amplify an initial enantiomeric imbalance and a plausible chemical pathway to transmit homochirality through stereoselective interactions, thereby enabling the emergence of a homochiral network spanning multiple classes of chiral biomolecules (*6*). Accordingly, a plausible scenario must couple a robust mechanism for achieving network-level homochirality with processes that could occur within the material and environmental constraints of early Earth (*7*).

Natural magnetic minerals—particularly magnetite (Fe_3O_4)—have recently emerged as promising candidates for facilitating chiral symmetry breaking and amplification en route to homochirality (*3*, *5*). Magnetite surfaces have been shown to act as strong chiral agents through the chiral-induced spin selectivity (CISS) effect (*16*), which couples electron spin polarization to molecular chirality, thereby enabling asymmetric processes controlled by electron spin (*17*–*20*). Recent experiments demonstrate that magnetized magnetite surfaces can serve as chiral nucleation templates, promoting the enantioselective crystallization of a central RNA precursor, ribose-aminooxazoline (RAO), and driving the system to full homochirality from initially racemic starting materials (*16*). Once an enantiomeric bias is established, enantiopure RAO crystals can, in turn, remagnetize the underlying magnetic surface, reinforcing magnetic asymmetry and enabling a feedback mechanism between surface magnetization and enantiomeric enrichment (*21*). Moreover, homochirality at the level of D-RAO can propagate to L-peptides and subsequently to key metabolites, supporting the emergence of a homochiral prebiotic reaction network (*4*, *6*, *22*, *23*).

While these findings demonstrate that magnetite can facilitate enantioselective processes, a crucial next step is to elucidate plausible formation pathways of magnetite under the prebiotic conditions of early Earth and to determine its magnetic properties in relation to spin-selective processes. Magnetite is known to form in various Fe(II)-rich environments and has been proposed as a primary precipitate from the Archean ocean (*24*–*27*). However, it remains important to investigate how environmental oxidants and solar UV irradiation contribute to its authigenic formation under the lacustrine conditions of early Earth's surface (Fig. 1), where selective prebiotic chemistry can generate chiral molecules with high yields, including RAO (*28*, *29*). Previous work (*30*) suggests that the concentration of nitrite on early Earth might have exceeded 1 μM in the oceans and could potentially have been much higher (near-millimolar) in ponds. Furthermore, to accurately simulate aqueous surface environments on early Earth, dissolved inorganic carbon (DIC) needs to be

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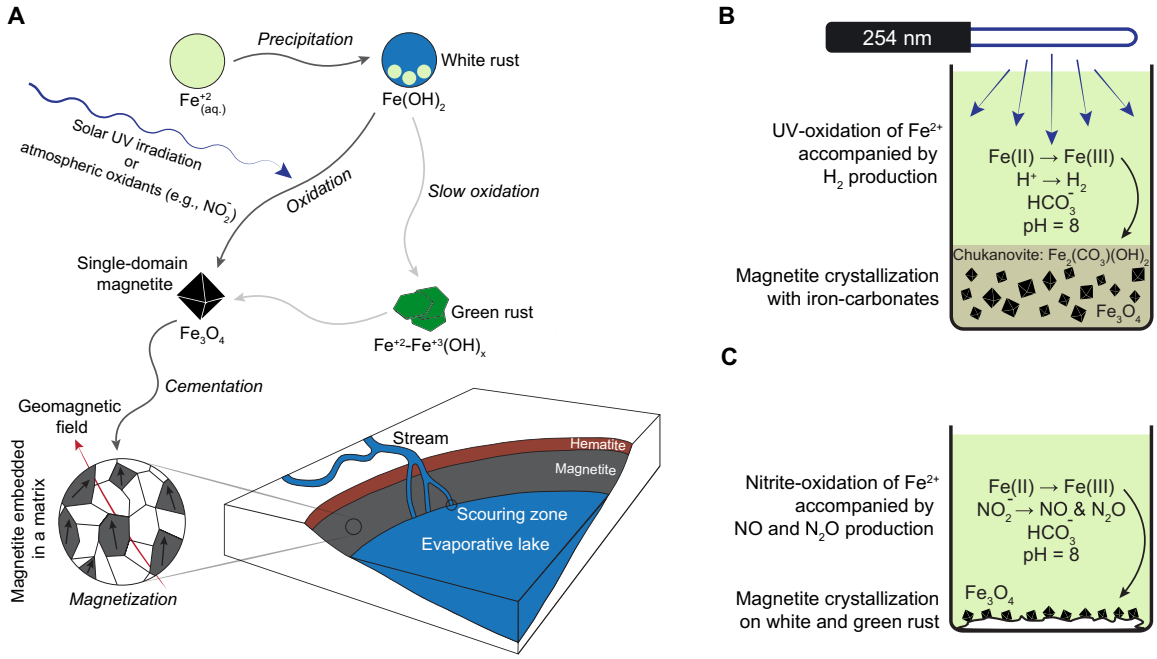


Fig. 1. Magnetite formation under prebiotically plausible conditions. (**A**) Envisioned evaporative lake setting on early Earth, where magnetite likely formed via authigenic precipitation facilitated either by solar irradiation or nitrogen oxide from the decomposition of abiotic nitrogen fixation products (*63*). As these magnetite particles form, they acquire a natural remanent magnetization under the geomagnetic field. (**B**) Experimental setup used to reproduce early Earth conditions and investigate magnetite formation under UV-facilitated synthesis (samples UV1 to UV3) and (**C**) via nitrite oxidation of Fe(II) (samples NUV8 and NUV9). Samples NUV5 to NUV7 were synthesized by mixing Fe(II) and Fe(III) with different proportions [10%/90%, 50%/50%, and 90%/10% of Fe(II)/Fe(III), respectively].

considered as well. While estimates for early Earth atmospheric CO_2 vary by several orders of magnitude (*31*), and postimpact scenarios are considered to transiently produce conditions with extremely low DIC (*32*, *33*), the presence of dissolved carbonate is known to strongly influence iron geochemistry by partitioning dissolved iron into iron-carbonate minerals. In this work, we study two plausible prebiotic magnetite formation mechanisms: the photo-oxidation of Fe(II) carbonates and their chemical oxidation by nitrite (Fig. 2).

Moreover, a key objective is to understand how distinct formation pathways influence the morphology and magnetic properties of the resulting magnetite, and how these properties, in turn, affect spin-selective interactions between chiral molecules and magnetite. It is well known that a magnetic material's response to external stimuli—applied fields, temperature changes, or remagnetizing agents such as chiral molecules—is fundamentally controlled by the domain state of its magnetic carriers (*34*). The domain state reflects the organization of magnetic moments within individual particles and depends primarily on grain size and shape. In the smallest particles, exchange interactions dominate the magnetic free energy, enforcing uniform moment alignment characteristic of a single-domain (SD) state (Fig. 3A, left). As particle size increases, this uniform configuration becomes energetically unfavorable, giving rise to nonuniform spin textures such as single-vortex (SV) or multivortex (MV) states, where magnetization curls around one or more vortex cores to reduce demagnetizing energy (Fig. 3A, center). With further growth, vortex structures give way to domain walls separating oppositely magnetized regions, defining the multidomain (MD) state (Fig. 3A, right). These configurations exhibit distinct magnetic properties—including coercivity, remanence, and magnetic stability—and thus behave variably under external magnetic influence. Understanding the distribution of domain states in prebiotically synthesized magnetite, as provided in our present manuscript, is therefore essential for evaluating whether chirality-induced spin interactions observed in laboratory settings could plausibly occur under early Earth conditions.

Connecting magnetic domain-level properties to molecular-scale, spin-selective chiral interactions requires an additional analysis. Magnetic domain properties of magnetite determine not only its bulk response to external fields but also the local spin environment at its surface. It is precisely at this interface that chiral molecules would interact with magnetization. Thus, any attempt to assess magnetite's potential as an enantioselective or symmetry-breaking agent must link macroscopic domain structures to the microscopic physics governing molecule-surface spin coupling (Fig. 4). Recent computational work on magnetic surfaces interacting with chiral molecules has largely operated at atomistic scales with ab initio modeling, where the quantum-mechanical nature of CISS-driven spin interactions can be explicitly treated (*35*). These models are valuable for estimating interaction strengths between chiral molecules and magnetic surfaces, but they remain restricted to small spin ensembles due to computational cost. As a result, they cannot assess how localized interactions between magnetic grains and a few spin-polarized chiral molecules might influence the magnetic domain state of an entire particle—a structure that emerges from the collective behavior of a macroscopic number of spins. However, determining whether and how chiral molecules can influence the magnetic structure of magnetite grains of realistic length and timescales is essential for evaluating magnetite's potential role as a chiral symmetry-breaking agent on prebiotic Earth. To address this, here we investigate the interaction of chiral molecules with real-sized prebiotic magnetite grains that we produce in our experiments, using micromagnetic modeling

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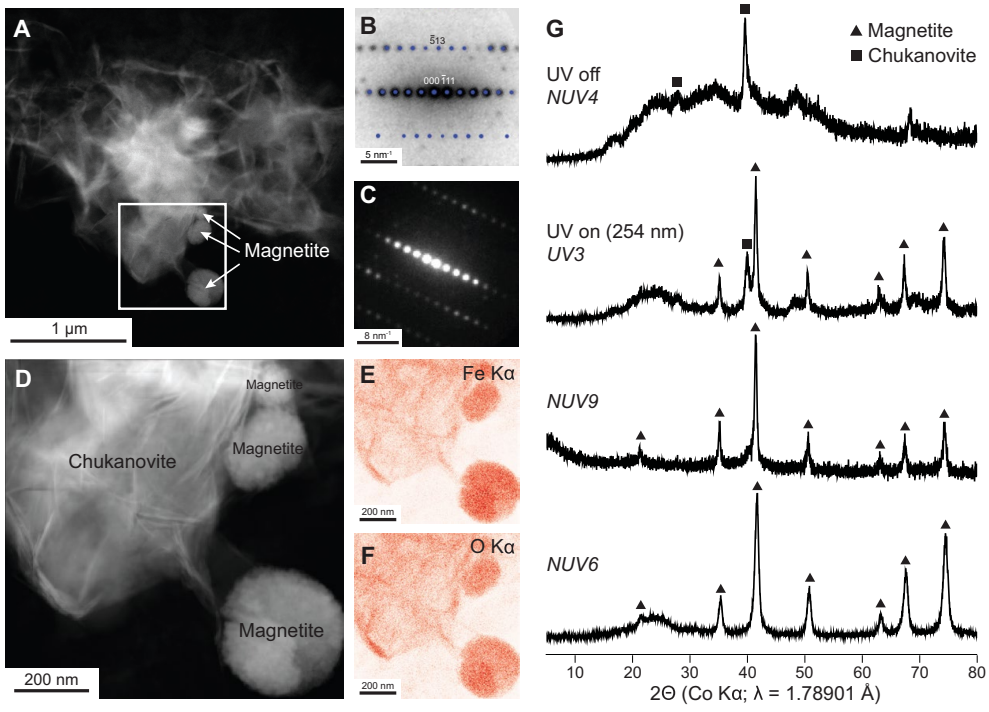


Fig. 2. Different prebiotic synthesis routes produce magnetite within a range of sizes and morphologies. (**A**) Transmission electron microscopy imaging from a UV-synthesized sample (UV2) showing nearly equidimensional, approximately 250 nm size magnetite particles embedded in acicular chukanovite (Chk) and goethite (Go) structures. (**B**) and (**C**) are electron diffraction data confirming magnetite structure and the detected crystal planes. (**D**) Zoomed-in region [white square in (A)] from where energy dispersive spectroscopy (EDS) chemical maps were taken, highlighting regions rich in (**E**) iron and (**F**) oxygen. (**G**) X-ray diffraction illustrating the impact of absence (UV off) and presence of Fe(II) oxidants (UV3, NUV9, and NUV6). Absence of oxidants completely prevents magnetite to form (NUV4); non-UV synthesized samples (NUV9 and NUV6) show clear magnetite peaks; a single evident chukanovite peak is also observed in both samples, while UV synthesis yields much clearer peaks for this phase (in addition to other iron oxide phases).

to evaluate the feasibility of spin exchange-driven remagnetization by chiral molecules (Fig. 4). Even for conservative estimates of the spin-exchange interaction, a substantial portion of magnetite grains can undergo irreversible magnetization by chiral molecules (Fig. 4E) and retain their remagnetized state over geological timescales (*36*). This process enables a persistent physical chiral bias of hemispheric-scale uniformity with a long lifetime for prebiotic chemistry in the geochemical environment, despite thermal fluctuations and geomagnetic reversals.

RESULTS

Magnetite synthesis under prebiotically plausible conditions

Magnetite was synthesized through four separate pathways. In all four cases, the experimental conditions (DIC = 20 mmol kg^{-1}) correspond to an equilibrium of approximately $10^{-2.3}$ bar of carbon dioxide in the atmosphere. This value is within the range of plausible estimates of the level of the pCO_2 for the Hadean Earth (*31*–*33*). In the first approach, $FeCl_2$ and $FeCl_3$ solutions were mixed in a 400-ml working matrix prepared (see Materials and Methods) in a 500-ml Pyrex beaker. After homogenization, the pH was adjusted to 8.0 with 1 mol kg^{-1} NaOH added drop wise. The mixture was stirred for 4 hours, transferred to a Duran bottle, sealed with a polypropylene cap, and incubated for 1 week. The resulting precipitate was collected by filtration through 0.22-μm membranes and dried in a low-humidity glovebox for 24 hours. Although effective, this method is unlikely to represent prebiotic conditions, as a rapid pH increase from 3 to 8 and the availability of $FeCl_3$ are both unrealistic for early Earth environments.

To better approximate prebiotic scenarios, a second method involved adding $FeCl_3$ dropwise into $FeCl_2$ while maintaining pH 8 via simultaneous NaOH titration. This procedure mimics the gradual formation of Fe(III) through slow oxidative processes. On early Earth, plausible oxidants include UV irradiation and nitrite (*30*, *37*). Therefore, in the third and fourth methods, Fe(II) solutions were partially oxidized either by UV exposure or by treatment with dilute sodium nitrite. Both oxidation pathways successfully produced magnetite (Fig. 2). The successful formation of magnetite under these relatively high dissolved CO_2 conditions—where competitive partitioning into carbonates is substantial—further strengthens the prebiotic plausibility of these synthesis pathways.

Magnetic properties of prebiotic magnetite

Bulk magnetic characterization measurements, including hysteresis, backfield curves, temperature dependence of magnetic susceptibility, and first-order reversal curves (FORCs), indicate that prebiotic synthesis of magnetite produces particles covering virtually the whole spectrum of domain states (Fig. 3). Although several samples contain mixed mineral assemblages, including minor goethite, chukanovite, and green rust (Fig. 2), the irreversible bulk magnetic signal is expected to be dominated by magnetite. Magnetite has a spontaneous magnetization of approximately 480 kA m^{-1}, whereas goethite is antiferromagnetic/canted antiferromagnetic and has a much weaker

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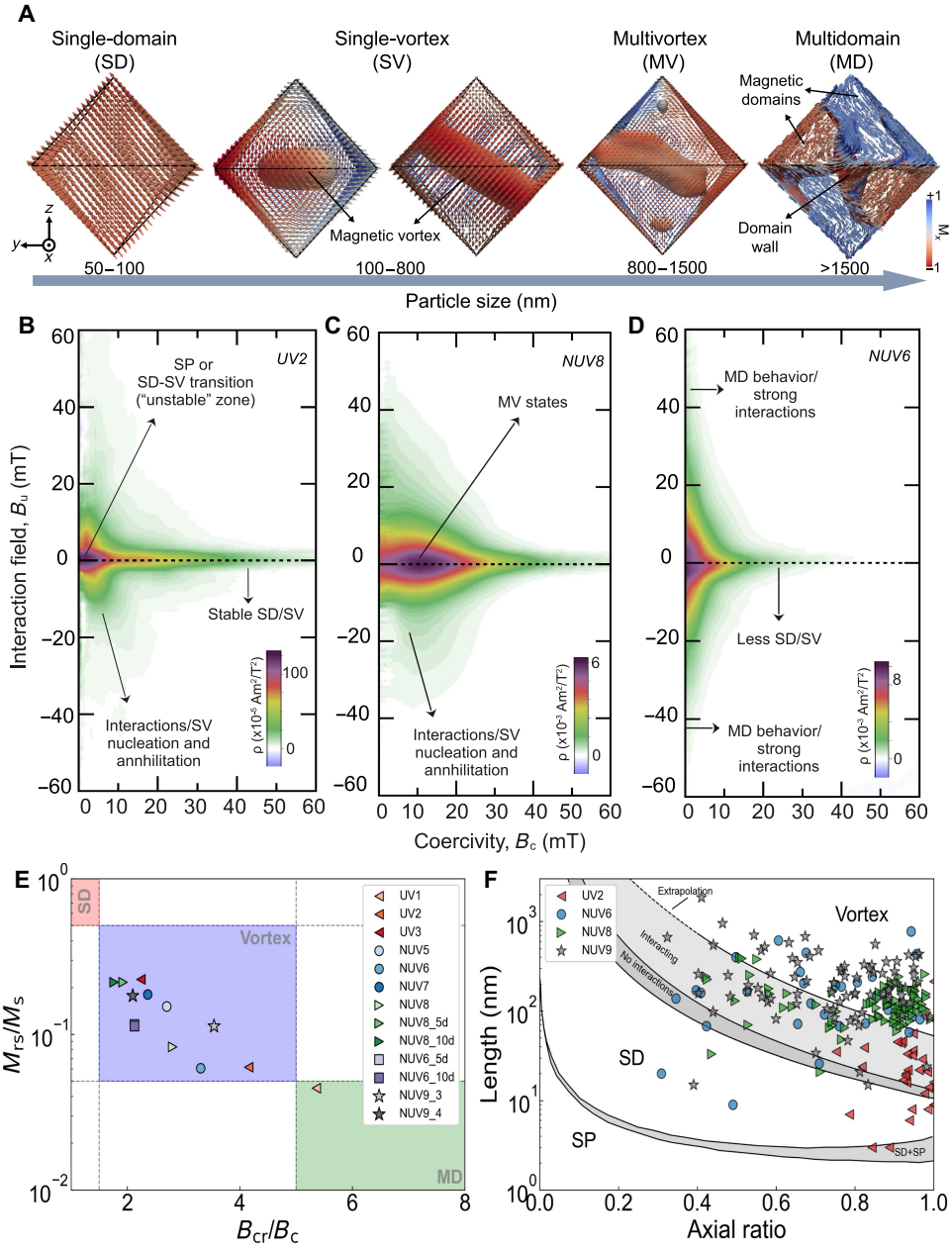


Fig. 3. Magnetic domain states in prebiotically plausible magnetite. (**A**) Different domain states obtained with micromagnetic modeling of individual magnetite particles with different sizes: single-domain (SD), single-vortex (SV), multivortex (MV), and multidomain (MD) states. (**B**) First-order reversal curves measurements for samples UV2 (UV-driven synthesis), (**C**) NUV8 (NO_2^--driven synthesis at pH 8.0), and (**D**) NUV6 [Fe(II) and Fe(III) mixture without pH controlling]. (**E**) Day plot (*64*) of the ratio between saturation remanent (M_{rs}) and saturation (M_s) magnetizations against the ratio between remanence coercivity (B_{cr}) and coercivity (B_c); light red, blue, and green regions are those where SD, SV, and MD particles fall within, respectively (*39*, *64*). (**F**) Butler-Banerjee plot delineating a two-dimensional size-shape range for SD and non-SD (i.e., SV, MV, and MD) particles of magnetite (*46*).

net magnetization, typically associated with high-coercivity remanence (*34*). Thus, even a small mass fraction of magnetite can dominate the ferrimagnetic component of the bulk magnetic response. Consistent with this interpretation, the hysteresis and FORC measurements do not show a pronounced high-coercivity component expected for a major goethite contribution. Other Fe(II)-bearing phases, such as green rust and chukanovite, are also not expected to produce the ferrimagnetic hysteresis or irreversible FORC structures used here to diagnose SD-SV-MV magnetite. Their contribution should appear primarily as a reversible, near-linear paramagnetic background in bulk magnetization curves. Because FORC distributions are obtained from the mixed second derivative of magnetization with respect to the applied field (*38*), such reversible linear terms do not generate FORC features and are strongly suppressed in the FORC representation. Therefore, the irreversible hysteresis and FORC features interpreted below primarily reflect the magnetite population.

In the Day plot (*39*), most samples cluster within the vortex region, suggesting that nonuniform magnetic structures dominate the

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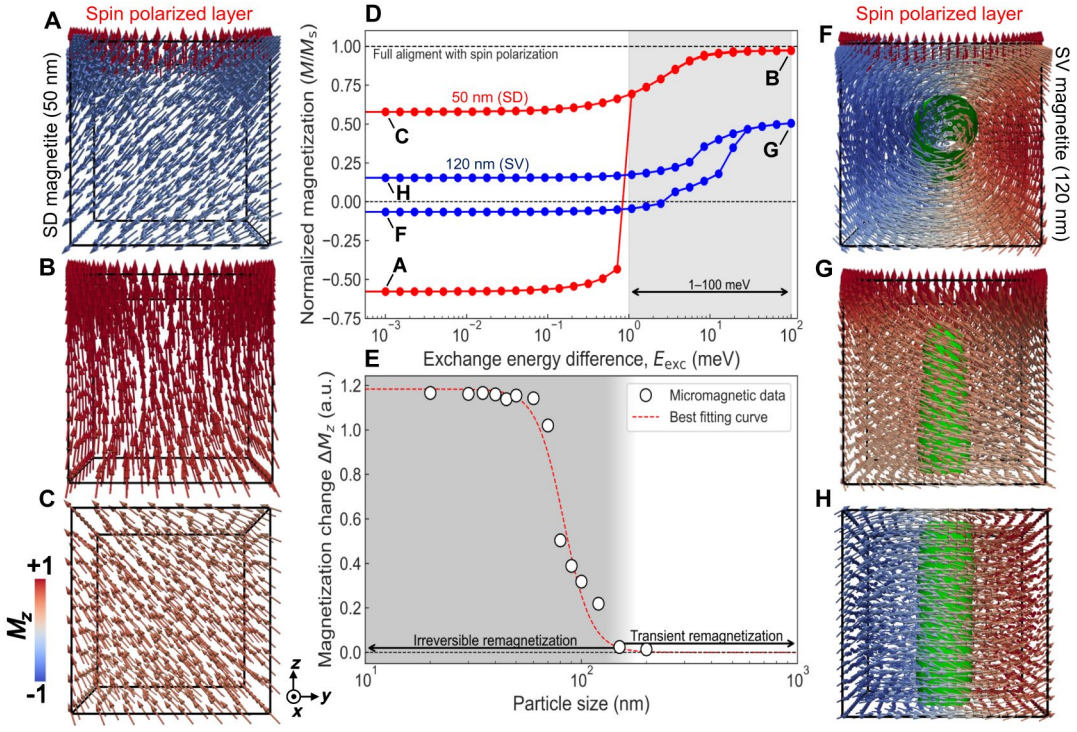


Fig. 4. Chiral molecules can irreversibly remagnetize single-domain and single-vortex states in magnetite particles. (**A** to **C**) Changes in the magnetic domain structure of a 50-nm, single-domain cube of magnetite caused by a layer of spin-polarized molecules [red arrows at the top (A), (B), (F), and (G)] as the strength of the exchange interaction (E_{exc}) between the spin-polarized layer and magnetite is varied from 0 to 100 meV and back from 100 meV down to 0. (A) Initial SD state aligned with the $[\bar{1}, \bar{1}, \bar{1}]$ direction; (B) intermediate remagnetized SD state aligned with the [0,0,1] direction (i.e., almost fully aligned with the layer's spin polarization); (C) final SD state aligned with the $[\bar{1}, \bar{1}, 1]$ direction after the exchange interaction strength is set back to zero. (**D**) Shows how the *z*-component of magnetization (i.e., the one along the fixed molecular spin polarization orientation) changes as a function of exchange interaction strength for a 50-nm–sized, SD magnetite particle (red curve) and a 120-nm–sized, SV particle (blue curve). (**E**) Magnetization change (defined as the difference between the *z*-components of the initial and final states after cyclically varying E_{exc}) as a function of grain size; particles with sizes below ~130 nm undergo irreversible remagnetization, while larger particles are only transiently remagnetized when exposed to the spin-polarized molecules. (**F** to **H**) Changes in the magnetic domain structure for a 120-nm, single-vortex magnetite particle: (F) initial SV aligned with [1,0,0] direction; (G) intermediate SV aligned with the [0,0,1] direction with a partially destroyed vortex as E_{exc} is at its maximum value, and (H) final [0,0,1] SV state with the vortex structure restored as the E_{exc} is set back to zero.

bulk magnetic signal (Fig. 3E). Nevertheless, the Day plot alone cannot resolve the coexistence of subpopulations with distinct domain structures, and thus FORC measurements were helpful to provide more detailed insights into their magnetic behavior. In UV-synthesized samples, FORC diagrams (Fig. 3B) show a pronounced central ridge along the coercivity axis (B_c) extending up to 60 mT, diagnostic of stable SD (~50 to 100 nm) and SV (≳100 nm) grains (*38*, *40*). The nearly symmetrical spread along the interaction-field axis ($|B_u| \lesssim 10$ mT) indicates weak to moderate magnetostatic coupling, while diagonal features reflect the nucleation and annihilation of vortex states on application of external fields (*38*, *40*). UV-synthesized samples also display a noticeable low-coercivity component near $B_c \approx 0$, pointing to the coexistence of thermally unstable, superparamagnetic (SP; <50 nm) grains and "critical-size" particles at the SD-SV threshold, where flattened magnetic energy landscapes produce very low stability (*36*, *41*). Contrastingly, samples produced via nitrite oxidation of ferrous iron display a large, central coercivity peak around ~13.7 mT (Fig. 3C), characteristic of particles adopting MV states (*42*), while samples produced from mixtures of Fe(II) and Fe(III) display near-zero coercivity peaks and signals spreading considerably along the interaction field axis, likely as an indication of strong interactions (*43*) and/or the presence of MD particles (*44*) (Fig. 3C).

Electron microscopy and energy-dispersive spectroscopy analyses further support the FORC interpretations by providing detailed morphological and compositional constraints on the synthetic magnetite. UV-driven synthesis produced a complex architecture of nearly equidimensional magnetite grains, ~15 to 200 nm in size (within the SP-SD-SV range; fig. S7), embedded within an acicular matrix containing chukanovite and goethite (Fig. 2, B to D and F, and fig. S6). Notably, these particle sizes fall precisely within the range where SV particles are expected to form according to high-resolution three-dimensional (3D) micromagnetic modeling (*36*, *40*, *45*), further supporting the interpretation of FORC measurements for UV-synthesized samples. On the other hand, imaging of non–UV-synthesized samples revealed larger aggregates, with individual magnetite particles reaching up to ~1.0 μm in length, consistent with the MV-MD domain states inferred from the magnetic data (fig. S3). Projecting the reconstructed size and shape distributions onto the Butler-Banerjee plot (Fig. 3F), which relates particle length to axial ratio and predicts domain states based on particle geometry (*46*, *47*), provides a quantitative view of these textural variations. In our dataset, particle dimensions range from a few tens of nanometers to more than 1 μm, with axial ratios between ~0.35 and 1.0, spanning nearly the entire size-shape space of the diagram but mostly

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concentrated on the vortex regions. Most UV-synthesized grains fall within the noninteracting zone, consistent with SV-like configurations (*48*), while the larger and more elongated grains formed through nitrite-mediated synthesis extend into the vortex field. Together, these observations confirm that both synthesis routes produce magnetite populations encompassing a broad size/shape distribution, but are mostly dominated by particles within the vortex size range.

Exchange-driven irreversible magnetization changes caused by chiral molecules

Having established that the synthesized magnetite populations are dominated by particles in the SD-SV-vortex size range, we next ask whether such particles are magnetically susceptible to interfacial exchange bias from CISS-active chiral molecules. This question is important because the magnetic response of isolated particles relevant to prebiotic settings cannot be inferred directly from thin-film CISS experiments: SD and vortex particles have distinct energy landscapes controlled by particle size, shape, magnetocrystalline anisotropy, and domain topology. We therefore used the 3D finite-element micromagnetic package MERRILL (*49*) to test whether a spin-polarized molecular layer, represented as a coarse-grained interfacial exchange term, can reorganize the magnetic structure of magnetite particles and produce irreversible changes in remanence (Fig. 4). Our micromagnetic results indicate that both SD (Fig. 4, A to D) and SV magnetite particles (Fig. 4, E to H) undergo remagnetization when exposed to spin-polarized materials. By cyclically varying the exchange interaction between spin-polarized molecules and magnetite particles from 0 to 100 meV [i.e., within the upper limits of ≈0.01 to 100 meV usually reported for such interactions (*21*)], our model was able to obtain exchange-driven "hysteresis" curves that confirm that the domain state changes in SD and SV states (up to ~100 to 130 nm) are irreversible, as indicated by the abrupt "jumps" observed in the lower branches of the "exchange-hysteresis" curves. As a consequence, once remagnetized by a spin-polarized material, SD and SV magnetite particle do not return to their initial state even after the removal of the molecules: An SD particle initially magnetized along the $[\bar{1}, \bar{1}, \bar{1}]$ direction (Fig. 4A) will end up magnetized in the $[\bar{1}, \bar{1}, 1]$ direction (Fig. 4C) when the exchange interaction energy is cyclically varied from 0 to 100 meV and back from 100 to 0 meV. Similarly, an initial SV state magnetized along the [1,0,0] direction (Fig. 4F) will lastly align with the [0,0,1] direction (Fig. 4H) as the interaction energy with the molecules is varied cyclically. Notably, both domain states end up with a larger contribution of their magnetization along the molecules' spin polarization (namely, their M_z component, Fig. 4, D and E). Therefore, based on our micromagnetic modeling, spin-polarized molecules are predicted to remagnetize SD and SV magnetite particles in a way to maximize their alignment with the molecules' spin polarization. This, coupled with the fact that the magnetization changes we observed with our models are irreversible, indicates that on early Earth settings, SD and SV magnetite particles are capable of providing a robust platform for developing and propagating a consistent chiral bias toward subsequent generations of homochiral crystals.

DISCUSSION

Relevance of magnetic domain states in chirality-induced magnetic interactions

The potential of magnetite to act as a natural symmetry-breaking agent for chiral molecules has recently been highlighted by experiments showing that MD magnetite thin films can induce enantiomeric excesses exceeding ~60% during the crystallization of racemic RAO, a key RNA precursor (*16*). Furthermore, RAO itself can trigger avalanche-like remagnetization of these films, propagating magnetic changes well beyond the initial contact zone (*21*). Together, these studies reveal a dual mechanism by which magnetite can (i) break chiral symmetry via its natural remanent magnetization acquired under Earth's magnetic field, and (ii) amplify this bias through feedback between homochiral molecules and magnetic surfaces.

Despite these advances, experimental demonstrations of chirality-induced magnetization changes have so far been confined to the ends of the magnetic size spectrum (Fig. 3A): nanometer-scale SP particles (≲10 to 30 nm) (*19*, *50*) and micrometer-scale MD thin films (≳1.5 to 3.0 μm) (*21*, *51*). The wide intermediate regime, spanning submicrometer- to micrometer-sized grains that adopt SV and MV states, has remained unexplored, even though such grains overwhelmingly dominate terrestrial igneous and sedimentary environments (*44*, *48*), as well as extraterrestrial materials (*52*, *53*).

Our results fill this critical gap: We show that under prebiotically plausible conditions, both UV-driven and nitrite-mediated magnetite synthesis pathways produce particles whose sizes and shapes fall mostly within the vortex state range. FORC analyses, electron microscopy imaging, and the reconstructed size-shape distributions (figs. S7 and S8) demonstrate that these grains form the majority of the magnetic assemblage, with SD-SV particles dominating the UV pathway and larger SV-MV grains characterizing the nitrite pathway. Thus, the geochemical processes likely responsible for magnetite formation on early Earth predominantly produce the domain states that have remained untested in previous CISS-related studies.

Irreversibility: The key for effectively propagating chiral bias

In our model, we consider a conservative exchange coupling between the spin-polarized molecular layer and the magnetite surface, in which the molecule-surface interaction is parameterized as an effective interfacial exchange bias, expressed as a per-interaction energy difference (ΔE) ranging from 0 to 100 meV (see Materials and Methods). Notably, AFM-based spin-exchange microscopy of chiral molecules on ferromagnetic films has identified that ΔE can be as large as 150 meV (*51*). Our simulations show that substantially smaller molecular exchange energies ($\Delta E \approx$ 0.1 to 30 meV; see Fig. 4D) are already sufficient to bias or reconfigure SD and vortex magnetite particles within the 20- to 170-nm size range. Consequently, the experimentally observed spin-exchange energy of ≈150 meV per interfacial interaction is more than sufficient, in principle, to generate the magnetic bias required to drive magnetization changes within this size range.

Further support stems from observations of chirality-induced magnetization switching in metallic thin films. Adsorption of a chiral self-assembled molecular layer has been shown to produce magnetization reversal in Co/Pt multilayers with an effective energy transfer on the order of milli–electron volts per electron (*17*). Similarly, molecular layers adsorbed on Ni thin films have been shown to induce magnetization switching equivalent to an applied field of ~0.4 T (*54*), while in magnetite thin films, a ~2.5-mT effective field was estimated to be caused by the adsorption of RAO molecules on it (*21*). Although these experiments use diverse materials and length scales, they consistently indicate that CISS-induced spin exchange at molecular interfaces can supply at least milli–electron volt–scale biases and, in some cases, field-equivalent perturbations of the order of milliteslas, far exceeding the geomagnetic field (tens of microteslas).

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The key finding obtained with our models is that both SD and SV magnetite grains undergo irreversible remagnetization when exposed to spin-polarized homochiral materials (Fig. 4). Critically, this means that once the exchange interaction between chiral molecules and the magnetic surface is cycled back to zero (i.e., mimicking when the crystals are removed or washed off), the magnetization does not return to its initial state. Instead, particles retain a remanence component aligned with the molecules' spin polarization. In SD grains, this results in a switch between crystallographic easy axes; in SV grains, it induces vortex-core realignment and reorientation of the net moment. In both cases, exchange-driven hysteresis demonstrates that even weak spin-polarized interactions can induce irreversible changes in realistic magnetite grains.

One should note, however, that our quasi-static, conservative micromagnetic approach does not explicitly include thermal activation. The purpose of the simulations is not to model the kinetics or timescales of magnetization switching, but rather to determine how a spin-polarized interfacial exchange bias reshapes the magnetic free-energy landscape and alters the relative stability and accessibility of competing domain states. Thermal fluctuations primarily affect the rate at which a particle can overcome energy barriers separating these states, but they do not change the existence or ordering of the minima themselves. In this sense, our results establish that the chiral exchange perturbation can drive the system across an irreversible boundary into a distinct remanent minimum, and the inclusion of thermally activated processes would then govern how rapidly such transitions occur in nature.

A natural concern is that, once the molecules are removed (i.e., once the interfacial coupling is cycled back to zero), the remagnetized states could gradually relax back toward the original configuration over time. For the vortex-bearing grains that dominate much of our size range, available estimate of relaxation time argue strongly against such decay on any relevant timescale: as shown by Nagy *et al.* (*36*), vortex states in magnetite grains larger than ~100 nm have relaxation times exceeding the age of the Solar System. This provides an important physical basis for interpreting the exchange-driven hysteresis observed in our simulations: Once the interfacial bias has reconfigured a vortex state, the resulting remanence is expected to remain effectively locked in even after the molecular layer is removed. For SD grains, thermal activation can, in principle, produce viscous relaxation when energy barriers are sufficiently small (i.e., for particles around the SP-SD threshold, 20 to 30 nm). Including thermal activation would not invalidate the existence of the biased remanent minima identified here; instead, it would be expected to facilitate relaxation toward the lower-energy, molecule-selected configurations over time when barriers permit.

The same reasoning implies that the resulting remanence should be robust to geomagnetic reversals. A reversal of the geomagnetic field changes the direction of a weak Zeeman bias (tens of microteslas), but this is orders of magnitude smaller than the coercive fields typical of SD-SV magnetite (millitesla scale, as reflected by the hysteresis parameters in table S1). Consequently, a reversing ambient field would be insufficient to reconfigure these remanent states unless additional processes (e.g., prolonged heating toward blocking conditions or exceptionally long-term viscous relaxation) substantially lower energy barriers. Therefore, once established, CISS-driven remanence components are expected to persist through polarity flips, allowing the magnetic asymmetry generated by homochiral molecular interactions to survive across repeated geomagnetic reversals.

This irreversibility is central to the plausibility of chiral feedback mechanisms under natural conditions. Unlike MD thin films, which require strong magnetic fields to induce a nonzero net magnetization, SD-SV grains formed in Fe(II)-rich lakes or sedimentary basins would already carry a weak but statistically coherent natural remanent magnetization acquired during growth or deposition under the geomagnetic field (*34*). Our results show that homochiral molecules such as RAO could reinforce and amplify this inherited bias by selectively remagnetizing particles in the direction of their own spin polarization. Because the induced changes are stable at zero effective field, the magnetic asymmetry produced by these interactions would persist across dissolution-recrystallization cycles and episodic mixing. Together, these findings reveal a coherent, geologically plausible pathway by which prebiotic magnetite could have contributed to both the initial symmetry breaking and subsequent amplification required to achieve homochirality.

In conclusion, our findings demonstrate that authigenic magnetite produced through prebiotically plausible pathways provides a robust geochemical basis for the emergence and persistence of biomolecular homochirality. We show that UV-driven and nitrite-mediated oxidation produce magnetite populations dominated by SV and MV grains. In addition, our 3D micromagnetic simulations demonstrate that these magnetite grains undergo irreversible, exchange-driven remagnetization when interacting with homochiral materials via the CISS effect. This irreversible exchange hysteresis provides a mechanism for reconfiguring the magnetization of magnetic minerals even for weak exchange energy differences arising from chiral molecule-magnetic surface interactions. This enables magnetite to store a persistent chiral bias relevant to biomolecular homochirality. When coupled with a geomagnetic field that induces uniform remanence across a planetary hemisphere, this process establishes a self-reinforcing feedback loop: Magnetite templates enantioselective crystallization, while the resulting enantiomerically enriched materials amplify and stabilize the mineral's magnetic bias. In summary, these results suggest that magnetite could have functioned as a persistent chiral agent enabling homochirality on prebiotic Earth.

MATERIALS AND METHODS

General method

Reagents and solvents were obtained from MP Biomedicals, Sigma-Aldrich. Reagents and solvents were used without further purification unless otherwise stated. A Metrohm 902 Titrando autotitrator combined with a Metrohm pH electrode was used to measure and adjust the pH to the desired value. All solutions were prepared in a Coy polymer glovebox filled with 5% H_2/95% N_2 gas and maintained under anoxic conditions by a pair of palladium catalysts and anhydrous $CaSO_4$ desiccants. All stock and working solutions were prepared using deionized water that was first deoxygenated by purging with O_2-free N_2 gas for at least 1 hour (*55*), then placed on a stir plate for 24 hours in the glovebox to allow the residual O_2 to escape and be removed by the catalysts. $FeCl_2$ and $FeCl_3$ stock solutions (0.5 mol kg^{-1}) were prepared by dissolving analytical-grade powders in the deoxygenated deionized water.

Working matrix preparation

$NaHCO_3$ solutions were freshly prepared for each experiment to ensure that the total DIC (20 mmol kg^{-1}) across experiments was not affected by the degassing of the solution over time. The stoichiometric ionic

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strength ($I = 0.1$ mol kg^{-1}) of all experiments was balanced by adding predetermined amounts of NaCl.

Magnetite synthesis by mixing $FeCl_2$ and $FeCl_3$ solution

The working matrix (400 ml) was prepared in a 500-ml Pyrex glass beaker. After the solution was homogenized by stirring, predetermined $FeCl_2$ and $FeCl_3$ solution was added (Table 1, entries 5, 6, and 7). The pH was adjusted to 8.0 by adding NaOH (1 mol kg^{-1}) solution dropwise. The resulting solution was stirred for 4 hours and then transferred to a Duran glass bottle and sealed with a polypropylene cap. After 1 week of incubation, the precipitate was collected by filtration through 0.22-μm membranes and dried in the low-humidity glovebox for 24 hours.

Magnetite synthesis by titrating $FeCl_2$ and $FeCl_3$ solution

The working matrix (400 ml) was prepared in a 500-ml Pyrex glass beaker and 3.6 ml of $FeCl_2$ stock solution was added (Table 1, entry 8). The pH was adjusted to 8.0 by adding NaOH (1 mol kg^{-1}) solution dropwise. After the solution was homogenized by stirring, 0.4 ml of $FeCl_3$ stock solution and NaOH (1 mol kg^{-1}) solution were added into the solution dropwise simultaneously to maintain the pH at 8.0. The resulting solution was stirred for 4 hours and then transferred to a Duran glass bottle and sealed with a polypropylene cap. After 1 week of incubation, the precipitate was collected by filtration through 0.22-μm membranes and dried in the low-humidity glovebox for 24 hours.

Magnetite synthesis by oxidizing green rust in $NaNO_2$ solution

The working matrix (400 ml) was prepared in a 500-ml Duran glass bottle. After the solution was homogenized by stirring, 4 ml of $FeCl_2$ (0.5 mmol kg^{-1}) solution was added. The pH was adjusted to 8.0 by adding NaOH (1 mol kg^{-1}) solution dropwise. After incubation for a week, the precipitate was filtered through 0.22-μm membranes and separated into three portions mixed with different concentrations of $NaNO_2$ solution (Table 1, entries 9 to 11). The resulting solutions were incubated inside the glovebox for another week. The precipitate was collected by filtration through 0.22-μm membranes and dried in the low-humidity glovebox for 24 hours.

Magnetite synthesis by irradiating $FeCl_2$ solution

The working matrix (400 ml) was prepared in a 500-ml Pyrex glass beaker. After the solution was homogenized by stirring, 4 ml of $FeCl_2$ (0.5 mmol kg^{-1}) solution was added (Table 1, entries 1 to 3, three replicates). The pH was adjusted to 8.0 by adding NaOH (1 mol kg^{-1}) solution dropwise. The resulting solution was stirred with UV irradiation for 4 hours and then transferred to a Duran glass bottle and sealed with a polypropylene cap. UV irradiation was performed using an Analytik Jena UVP Pen-Ray mercury lamp (90-0004-01, 25 mA) placed above the solution. A control experiment has been run in parallel without UV irradiation (Table 1, entry 4). After 1 week of incubation, the precipitate was collected by filtration through 0.22-μm membranes and dried in the low-humidity glovebox for 24 hours.

Rock magnetic measurements

Mass-specific magnetic susceptibility was measured at 976 Hz using an Agico Multifunction Kappabridge susceptibility meter (model MFK1-FA), with an applied field strength of 200 A/m. To minimize measurement uncertainty, each sample was measured five times, and the average susceptibility values were used. Magnetic hysteresis loops, backfield demagnetization curves, and FORCs were measured using a Lakeshore PMC MicroMag 2900 alternating gradient field magnetometer. Hysteresis loops were acquired in discrete mode with a maximum applied field of 1 T and an increment of 2 mT. Hysteresis parameters (saturation magnetization M_s, remanent magnetization M_{rs}, and coercive force B_c) were determined after correcting for the high-field slope (>700 mT) to remove diamagnetic and paramagnetic contributions. The coercivity of remanence B_{cr} was obtained by progressively demagnetizing a saturated sample in a reverse field down to −1 T. FORC diagrams were acquired at maximum applied fields of 1 T with 1.02-mT field increments and processed using FORCinel software (version 3.08) with smoothing parameters summarized in table S3 in the Supplementary Materials (*56*, *57*). Before processing, the lower branch of the hysteresis loop was subtracted from each FORC to generate a set of "difference FORCs." Although this subtraction does not alter the FORC distribution, it significantly reduces common processing artifacts and removes sigmoidal contributions near the remanence diagonal, which cannot be adequately captured

Table 1. Experimental conditions for magnetite synthesis.

Entry	Sample no.	$FeCl_2$ (mmol kg^{-1})	$FeCl_3$ (mmol kg^{-1})	$NaNO_2$ (mmol kg^{-1})	Irradiation time (hours)	Incubation time (days)
1	UV1	5	0	0	4	7
2	UV2	5	0	0	4	7
3	UV3	5	0	0	4	7
4	NUV4	5	0	0	0	7
5	NUV5	4.5	0.5	0	0	7
6	NUV6	2.5	2.5	0	0	7
7	NUV7	0.5	4.5	0	0	7
8	NUV8	4.5	0.5	0	0	7
9	NUV9_1	5	0	0	0	7
10	NUV9_3	5	0	0.02	0	7
11	NUV9_4	5	0	0.1	0	7

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by the second-order polynomial function used to smooth the FORC surface (*57*).

Electron microscopy and energy-dispersive spectroscopy

Scanning electron microscopy (SEM) images were acquired using a FEI Quanta 650 FEG SEM in high-vacuum mode. Backscattered electron (BSE) SEM images were acquired at accelerating voltages of 5 to 10 kV, while energy-dispersive x-ray spectroscopy elemental mapping was conducted at 5 to 15 kV. High-resolution BSE-SEM images (1 nm/pixel) were captured using AZtec software (version 6.2). For transmission electron microscopy (TEM) measurements, the powder was dispersed onto a holey carbon film supported on a 200-mesh copper grid from EM Resolutions by suspending the powder in water, pipetting the suspension onto the grid, and allowing the water to evaporate. The sample was imaged using scanning TEM (STEM) with an aberration-corrected Thermo Fisher Scientific Spectra 300, operated with an accelerating voltage of 300 kV, in the Department of Materials Science and Metallurgy at the University of Cambridge. STEM images and EDS datasets were acquired simultaneously, using a beam current of approximately 100 pA and a convergence angle of 23 mrad, with a dwell time of 1 ms. EDS datasets were collected with the Super-X detector, consisting of four silicon drift detectors, and STEM images were collected with the Panther segmented STEM detector system, with a HAADF inner angle of 50 mrad. STEM and EDS images were analyzed using Hyperspy (*58*). Microscopy images have then been used to extract particle size and shape distribution using image segmentation with the software ImageJ (*59*).

Scanning electron diffraction

Scanning electron diffraction (SED) datasets were acquired also with the Thermo Fisher Scientific Spectra 300 and accelerating voltage as in the scanning TEM, with a 0.2-mrad convergence angle, a probe current of approximately 30 pA, and a dwell time of 1 ms. SED patterns were recorded on a Quantum Detectors MerlinEM direct electron detector.

SED datasets were analyzed using pxyem (*60*). The reciprocal space lengths in the diffraction patterns were calibrated based on patterns collected under the same conditions from AuPt nanoparticles. After centering the direct beam, a potential magnetite particle was identified based on its morphology, and the composition was determined by EDS. Kinematical diffraction patterns were simulated using the Single Crystal 5 software and compared with the summed diffraction patterns from the magnetite particle.

X-ray diffraction

Powder x-ray diffraction spectra were acquired using a Malvern Panalytical Empyrean instrument with a Co K source (1.78901 Å) operated at 45.0 kV and 40.0 mA, scanned between 2° and 80° at a step size of 0.026° and a total scan time of approximately 11 min.

Micromagnetic modeling

Micromagnetic simulations were performed to quantify how a CISS-active, spin-polarized molecular layer can bias the domain-state energetics of magnetite particles. In micromagnetics, the magnetization is represented by a continuous unit-vector field $\mathbf{m}(\mathbf{r})$, and quasi-static configurations are obtained by minimizing the total magnetic free energy, including exchange, magnetostatic (demagnetizing), magnetocrystalline anisotropy, and Zeeman terms (*61*). Calculations were carried out using MERRILL, an open-source finite-element micromagnetic software suitable for natural magnetic minerals (*49*). Room-temperature magnetite parameters were fixed to experimentally constrained values: exchange stiffness $A_{\mathrm{mag}} = 1.33 \times 10^{-11}$ J m^{-1}, magnetocrystalline anisotropy $K_1 = -1.24 \times 10^4$ J m^{-3}, and spontaneous magnetization $M_s = 480$ kA m^{-1} (*49*).

Effective interfacial coupling to a chiral molecular layer

To incorporate the magnetic influence of a chiral molecular layer generated via CISS, we adopt a coarse-grained interfacial description: The molecular layer is not treated as a magnetic medium with its own internal exchange or micromagnetic dynamics; instead, its net effect is represented as an interfacial energetic bias that favors one relative alignment between the magnetite surface magnetization and an effective molecular polarization axis $\mathbf{m}_{\mathrm{mol}}$. This formulation does not assume that the underlying microscopic mechanism is conservative or specify a particular nonequilibrium torque pathway; rather, it tests whether experimentally and electronically motivated interfacial energy scales are sufficient to reshape the particle's quasi-static free-energy landscape and thereby modify domain-state stability and reversal pathways.

The mapping between the microscopic molecule-surface interaction and the continuum micromagnetic coupling is derived in the Supplementary Materials. Briefly, a surface density n_s of active molecule-surface interactions, each with an antiparallel-parallel energy difference ΔE_{pair}, produces an surface energetic bias

$$\Delta e_{\mathrm{mol}} = n_s \Delta E_{\mathrm{pair}} \tag{1}$$

In the micromagnetic model, this surface bias was implemented by inserting a thin intermediate layer (IL) of thickness d between the magnetite surface and a pinned molecular layer with spin polarization direction $\mathbf{p}$. The IL was assigned $M_s = 0$ and $K_1 = 0$, so that it does not contribute to magnetostatic or anisotropy energy. Its only role is to mediate an exchange coupling between the magnetite surface and the fixed molecular polarization direction. Under the thin-layer approximation, the corresponding micromagnetic interfacial energy density is

$$e_{\mathrm{int}} \simeq \frac{2A_{\mathrm{ex}}^{\mathrm{IL}}}{d}(1 - \mathbf{m}\cdot\mathbf{p}) \tag{2}$$

which gives an antiparallel-parallel areal energy difference

$$\Delta e_{\mathrm{int}} = \frac{4A_{\mathrm{ex}}^{\mathrm{IL}}}{d} \tag{3}$$

Equating the molecular and micromagnetic interfacial energy differences, $\Delta e_{\mathrm{mol}} = \Delta e_{\mathrm{int}}$, gives

$$A_{\mathrm{ex}}^{\mathrm{IL}} = \frac{n_s \Delta E_{\mathrm{pair}} d}{4} \tag{4}$$

Here, ΔE_{pair} is the effective per-interaction energetic preference between opposite relative alignments of the molecular spin-polarization and the magnetite surface bias. The sign of the preferred alignment is encoded in the choice of $\mathbf{p}$; reversing the molecular handedness or the preferred spin alignment is represented by replacing $\mathbf{p}$ with $-\mathbf{p}$. The parameter d is not interpreted as the physical molecule-surface separation. It is the numerical thickness over which the interfacial areal energy is distributed in the finite-element mesh.

We explored ΔE_{pair} from 0.01 to 100 meV, spanning weak to strong CISS-mediated interfacial coupling while avoiding reliance on a single literature value. For a molecular surface density $n_s = 2 \times 10^{18}$ m^{-2}

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(*17*, *62*) and an intermediate-layer thickness $d = 1.0$ nm, Eq. 4 gives $A_{ex}^{IL} = 8 \times 10^{-17}$ to 8×10^{-12} J m^{-1}, corresponding to approximately 6×10^{-4} to 60% of the internal exchange stiffness of magnetite. Because n_s and ΔE_{pair} enter only through the product $n_s \Delta E_{pair}$, lower molecular coverage is equivalent to shifting the effective ΔE_{pair} scale downward.

The implementation of Eq. 4 was validated in the Supplementary Materials by prescribing ΔE_{pair}, calculating A_{ex}^{IL}, and measuring the additional micromagnetic exchange energy recovered from the multiphase model. For the particle sizes and thin ILs used here, the recovered micromagnetic energy difference closely follows the prescribed ΔE_{pair}, confirming that the intermediate-layer implementation reproduces the intended interfacial bias. Deviations occur only for very small particles when artificially large intermediate-layer thicknesses are used, where the thin-layer approximation begins to break down (fig. S12).

For comparison with experiments, we also define an equivalent bias field B_{eff} as the uniform applied field that would reproduce the same change in magnetization as the interfacial coupling. This field is only a diagnostic quantity: The molecular effect is interfacial, whereas a true applied magnetic field acts throughout the particle volume. For the particle sizes explored here and for the adopted n_s and d, $\Delta E_{pair} \sim 0.1$ to 10 meV corresponds to equivalent bias fields of order 0.1 to 30 mT, comparable to millitesla-scale perturbations inferred from adsorption-driven magnetic experiments (*4*, *54*).

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S12
Tables S1 and S2
References

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Acknowledgments: S.F.O. and D.D.S. acknowledge the Harvard Origins of Life Initiative. S.F.O. further acknowledges the Fellows of King's College Cambridge for their hospitality. T.G. acknowledges the Wolfson Electron Microscopy suite for the use of the TF Spectra 300. Last, we are grateful to R. Danisi and I. Buisman for assistance with electron microscope imaging, and to E. Mansbach for insightful discussions. **Funding:** We thank the Origins Federation for funding and its members for fruitful discussions. S.F.O. acknowledges support from the Kavli-Laukien fellowship program, Caltech startup funds, and the William H. Hurt Scholarship Program. R.J.H., N.J.T., and J.A.P.M.D. acknowledge the Leverhulme Centre for Life in the Universe (LCLU) funding (LBAG/329), with additional LCLU support for N.J.T. and Z.L. (RC-2021-032). Part of this work was performed by J.A.P.M.D. as a visiting researcher at the Institute for Rock Magnetism (IRM) at the University of Minnesota, which is supported by the National Science Foundation (NSF EAR-2153786) and the University of Minnesota. **Author contributions:** S.F.O., N.J.T., R.J.H., J.A.P.M.D., and D.D.S. conceived the study. Z.L., N.J.T., R.J.H., C.Z.J., S.F.O., J.A.P.M.D., and T.G. developed the methodology. Z.L., S.F.O., J.A.P.M.D., and T.G. conducted the investigation, curated and analyzed the data, performed validation and formal analysis, and prepared the visualizations. S.F.O., Z.L., and J.A.P.M.D. wrote the original draft, and Z.L., N.J.T., R.J.H., S.F.O., D.D.S., J.A.P.M.D., and T.G. contributed to reviewing and editing the manuscript. Z.L., N.J.T., R.J.H., C.Z.J., D.D.S., and T.G. provided resources. N.J.T., R.J.H., S.F.O., and D.D.S. secured funding and contributed to supervision and project administration. J.A.P.M.D. developed the software. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All data and code needed to evaluate and reproduce the results in the paper are present in the paper and/or the Supplementary Materials. The raw data and computational materials have been deposited in a Zenodo repository available at https://zenodo.org/records/21790036. The repository includes raw and processed rock-magnetic, microscopy, diffraction, and micromagnetic modeling data. All micromagnetic results reported here were generated using the open-source micromagnetic modeling code MERRILL (*49*). A complete guide to the installation and use of MERRILL is available at https://blogs.ed.ac.uk/rockmag/. This study did not generate new materials or code.

Submitted 11 February 2026
Accepted 2 October 2026
Published 6 November 2026
10.1126/sciadv.aeg3954

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Sci. Adv. **12** (45), eaeg3954. DOI: 10.1126/sciadv.aeg3954

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